Local goodness-of-fit measures for neural decoding

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Abstract

State-space models are widely used in neuroscience to infer latent states from observations, but validating them remains difficult. Interpretable models are necessarily simplified relative to the data, and determining the level of complexity needed to describe latent variables accurately is challenging. This challenge results from two main factors. First, because latent variables are inherently unobservable, decoding quality cannot be validated directly against observed data. Second, prediction errors caused by model misspecification can be difficult to interpret. We therefore developed metrics designed to identify localized periods when individual observations conflict with the model's accumulated estimate. Each metric assesses the consistency between the information contained in an individual observation and the information summarized by a state distribution over a chosen window of data. In the analyses here, we use the one-step predictive distribution, which carries forward information from all observations through the preceding time bin; more generally, the comparison could use filtering or smoothing distributions defined over other windows. The metrics fall into three categories: 1) measures of overlap between highest probability-density regions of the state distributions; 2) predictive checks based on the model-implied predictive probabilities of individual observations; and 3) the Kullback–Leibler (KL) divergence, included as a familiar baseline that quantifies difference rather than consistency. We applied these metrics to simulated data and to real rat hippocampal neural data, demonstrating their ability to identify periods of model misspecification and to provide interpretable feedback for model improvement. Together, the metrics provide complementary tools for neuroscientists to assess the local goodness-of-fit of state-space models, enabling more informed decisions about model refinement and interpretation of neural decoding results.

Keywords: Neural decoding, State-space models, Goodness-of-fit

Introduction

Internal brain states related to memory, emotion, and attention play central roles in behavior. They arise from a time-varying interplay between the external world and internal processes and are not directly observable, making them difficult to characterize.

Recordings of neural activity, including spike trains, calcium signals, and voltage traces, provide experimental measurements influenced by these latent states. State space models make it possible to relate these measurements to estimates of the latent states themselves. Specifically, these models provide a method for relating complex, high-dimensional neural activity to estimates of low-dimensional latent states, and have provided insights into neural representations and dynamics associated with taste [1], spatial location [2], and aggression [3].

State space models aggregate information about the latent states over time. They combine the information from the data at the current time (the normalized likelihood distribution) with the accumulated information used to predict the latent state (the prediction distribution). This powerful combination enables robust estimation of the latent state, effectively balancing current information with the accumulated history of the system and prior knowledge.

Despite the wide use of state space models in neuroscience, methods to validate these models remain underdeveloped and underused. One reason is that latent states are inherently difficult to evaluate [4]. In the absence of ground truth, model validity can only be inferred indirectly, such as by comparing relative likelihoods under multiple models, the quality of reconstructed signals, or performance on downstream tasks. Neural data are also dynamic, so model fit may vary substantially over time. Compounding this problem is that common methods of validating models, such as cross-validation and "shuffle" permutation tests, are holistic evaluations of the model — they tell neuroscientists whether or not the model generalizes well or is different from a simplified model with less utility — but they fail to indicate when the model fits poorly or which components are responsible. Consequently, these methods can reject models that are intentionally simplified yet scientifically meaningful and offer little guidance for targeted improvement of the model. For example, analyses to decode rat movements from hippocampal place cells typically ignore known features of the rat's movement (such as momentum) and known features of neurons (such as refractoriness and bursting). These omissions can produce detectable model mismatch even when the resulting decoding trajectories remain scientifically informative [5–7].

State space models already contain rich diagnostic information in the form of distributions used to estimate the state process at each time, including the normalized likelihood, which reflects information from only the current neural observations, and the prediction distribution, which aggregates information from all past observations based on the state space model structure. One potential solution for more targeted local diagnostic information would be to directly compare these distributions. We do not expect these two distributions to be identical. The prediction distribution aggregates information from many past spikes; the likelihood reflects only the current observation. The two distributions therefore carry different amounts of information about the state and will typically have different shapes and spreads. What matters for model assessment is not whether the distributions differ, but whether they are consistent, meaning that they concentrate their probability mass over overlapping regions of the state space rather than disjoint ones. When these distributions provide inconsistent information about the state, the mismatch can reveal where, when, and potentially how the model fails to capture the structure of the data. Examining these discrepancies could therefore provide a direct, interpretable measure of model–data agreement—analogous to a local, time-resolved goodness-of-fit test.

To address these gaps, we develop and validate a suite of local goodness-of-fit methods, each built on this comparison between the prediction distribution and the likelihood. The most familiar way to compare two distributions is to measure their divergence, and therefore we include KL divergence as a natural point of reference. However, divergence measures quantify how much two distributions differ, not whether they are consistent, so they flag any mismatch, including broad prediction distributions or narrow likelihoods even when the models capture the data structure well. We therefore develop two consistency-focused measures, highest probability-density (HPD) region overlap and a predictive check measure, and show that they flag genuine model–data inconsistency and provide actionable feedback to neuroscientists. Using simulated and real hippocampal data, we demonstrate how each metric responds to different types of misspecification and argue that consistency is the right target for local goodness-of-fit assessment. This approach offers a path for neuroscientists to move from binary model rejection (is this model good or bad?) toward an iterative, diagnostic approach to building interpretable theories of neural dynamics.

Methods

State-space models

The state-space paradigm is particularly well suited for neural data analysis because many neuroscience experiments focus on internal brain processes that are not directly observable. For example, hippocampal replay of spatial trajectories has been hypothesized to be related to mental exploration of space, but the trajectories being replayed are not directly observable by experimentalists [6]. In such cases, state-space models are used to simultaneously estimate the unobserved internal representations and their relation to neural activity patterns.

For a discrete set of time points, $t_0, \ldots, t_K$, we define an unobserved state process x_k, which influences a neural observation process y_k at each time t_k. We define a state space model by combining a state transition distribution $p(x_k \mid x_{k-1})$, which defines how the state evolves in time, with an observation likelihood $p(y_k \mid x_k)$, which defines the relationship between the neural activity and the state. To simplify the problem, we assume that the state transition distribution depends only on the most recent past value of the state and that the observation likelihood depends only on the current state value. Fig 1a shows a schematic of this state-space model structure.

This state-space model structure allows for iterative estimation of the state process at each time step based on all the neural data up to that time (Fig 1b). If the distribution of the state at time t_{k-1} has been computed, we can convolve it with the state transition distribution to obtain a one-step prediction distribution, $p(x_k \mid y_{1:k-1}) = \int_{x_{k-1}} p(x_k \mid x_{k-1}) \cdot p(x_{k-1} \mid y_{1:k-1})\, dx_{k-1}$. This predictive distribution combines the information from all previous observations to compute the distribution of the state at the next time step. We update this predictive distribution with the information from the most recent observation by multiplying by the observation likelihood to compute the posterior distribution of the state given all of the data up to the current time step: $p(x_k \mid y_{1:k}) \propto p(y_k \mid x_k) \cdot p(x_k \mid y_{1:k-1})$.

Common approaches for assessing the goodness-of-fit of state-space models include examining the cross-validated error measures in the state process when the state is observable [8], cross-validated error measures in the observation process such as the mean-squared prediction error [9], and the distribution of prediction residuals [10]. Although these approaches can be useful, they may be difficult to interpret for neural data because intentionally simplified state-space models often exhibit some detectable lack of fit even when they remain scientifically useful.

Here, we take a different approach that relies on a set of local goodness-of-fit measures. Each measure compares the one-step predictive distribution with either the normalized observation likelihood as a function of state or the observation itself. These measures therefore consider the consistency between information accumulated through the state-space model over past spiking

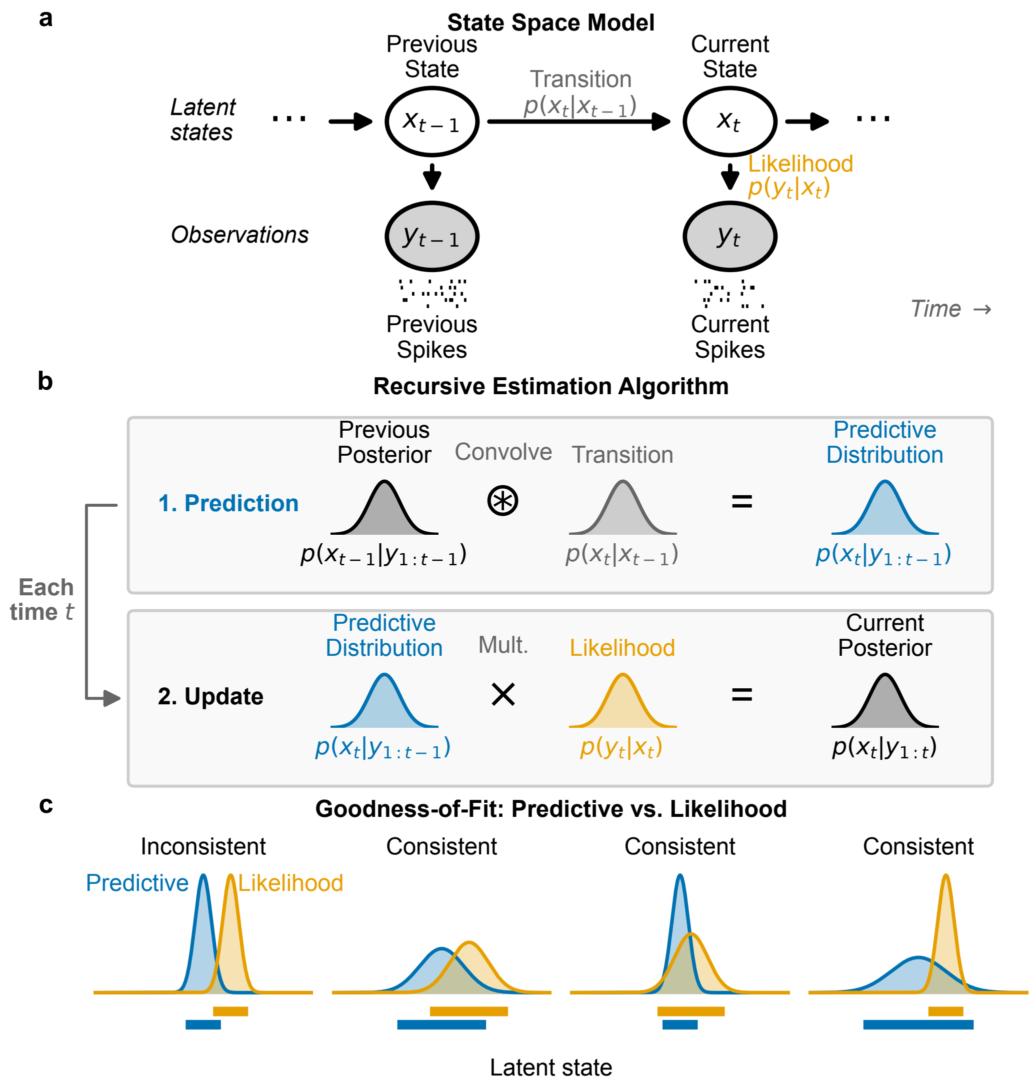

Fig 1. State-space model and local goodness-of-fit measures. **(a)** Graphical model depiction of state-space model includes state transition model, $p(x_k \mid x_{k-1})$, and observation likelihood $p(y_k \mid x_k)$. **(b)** These model components are sufficient to compute an iterative expression for the posterior filter distribution at time t_k, $p(x_k \mid y_{1:k})$ from the distribution at the previous time step. This involves first convolving the previous posterior with the state transition distribution to compute the one-step predictive distribution, $p(x_k \mid y_{1:k-1})$, and then updating the predictive distribution with information from the most recent spiking by multiplying by the observation likelihood. **(c)** For each new spike observation, we compute the local goodness-of-fit by assessing the consistency of information between the predictive distribution and likelihood. We consider the distributions inconsistent when they assign high probability to mostly disjoint subsets of the state-space. When the regions with high probability overlap, the distributions are considered consistent, even if the distributions differ in their central tendency or certainty.

and the information from the most recent spiking activity. Critically, we do not expect these distributions to align perfectly since they likely contain different amounts of information about the state process. Instead, we use the operational notion of consistency illustrated in Fig 1c. We consider the distributions inconsistent when their probability masses are concentrated in mostly non-overlapping regions of the state space. We consider them consistent when their high-probability regions largely overlap, even if their central tendencies differ. We also regard the distributions as consistent when a broad high-probability region of one distribution contains a substantially narrower region of the other. This situation may arise frequently because the predictive distribution contains information accumulated from many previous spikes.

In the analyses here, "local" refers to evaluating the diagnostic at an individual spike, not to truncating the past to a fixed-width window. The one-step predictive distribution summarizes all observations through the preceding time bin via the recursive state-space filter. More generally, the same comparisons could use a filtering or smoothing distribution defined over another chosen window, including a window containing both past and future observations.

Local goodness-of-fit measures

In this section, we introduce three classes of metrics designed to characterize the local goodness-of-fit of the state-space model.

All three metrics are evaluated at the level of individual spike events. Let $y_{k,j}$ denote the mark of the jth spike event in time bin k: the neuron identity for a spike-sorted model or the waveform features for a clusterless model. For the spike-sorted models analyzed here, HPD overlap and KL divergence compare the one-step predictive density

$$P_k(x_k) = p(x_k \mid y_{1:k-1})$$

with the normalized one-spike likelihood

$$Q_{k,j}(x_k) = \frac{\text{Poisson}\Big(1;\ \lambda_{y_{k,j}}(x_k)\Big)}{\int \text{Poisson}\Big(1;\ \lambda_{y_{k,j}}(u)\Big)\ du}, \tag{1}$$

where $\lambda_c(x_k)$ is the model-implied expected spike count for neuron c in time bin k at state x_k. If $r_c(x_k)$ denotes the corresponding conditional intensity and Δt is the bin width, then $\lambda_c(x_k) = r_c(x_k)\Delta t$; throughout, we use λ_c for this per-bin Poisson mean. Thus, $Q_{k,j}$ is the likelihood contributed by the observed spike event alone, rather than the joint likelihood of all spike counts in the bin. If several spikes occur in one bin, each event is evaluated separately against the same one-step predictive density P_k. If the observed neuron's expected count is zero at every evaluated state, the denominator in Eq. (1) vanishes; as in the implementation, we then define $Q_{k,j}$ to be uniform over the state grid because the event carries no positional information under this likelihood. For a clusterless model, the analogous $Q_{k,j}$ is obtained by normalizing the single-event waveform-mark likelihood over the state space.

Overlap between HPD regions

One way to assess the consistency between the predictive distribution and the observation likelihood is to examine the overlap between their highest probability-density (HPD) regions. We refer to the resulting measure as HPD overlap. A substantial intersection indicates that the distributions assign high density to some of the same state values. Minimal or no overlap indicates disagreement under this criterion. This metric provides an interpretable geometric summary of local consistency between the distributions.

We define the overlap between the HPD regions of the predictive distribution and the normalized one-spike likelihood as

$$\text{HPDOverlap}(P_k, Q_{k,j}) = \frac{\left|\text{HPD}_{P_k} \cap \text{HPD}_{Q_{k,j}}\right|}{\min\left\{\left|\text{HPD}_{P_k}\right|,\ \left|\text{HPD}_{Q_{k,j}}\right|\right\}}, \tag{2}$$

where HPD_{P_k} and $\text{HPD}_{Q_{k,j}}$ are the highest probability-density regions of the predictive density and the normalized one-spike likelihood, respectively, and $|\cdot|$ denotes the volume of the region. The numerator is the size of the overlap between these regions and the denominator is the size of the smaller of the two HPD regions. This normalization ensures that the resulting value is in the range $[0, 1]$, with 0 indicating no overlap, and 1 indicating that one region is entirely contained within the other. For example, if the likelihood region is wider and therefore less informative than the predictive region but completely contains it then the HPD ratio will be 1, suggesting that the two distributions are consistent under this criterion. Conversely, if the ratio is close to 0, it indicates little or no overlap, suggesting significant differences between the range of values for the state that would be inferred from these two distributions. Throughout, we compute 95% HPD regions.

Fig 2a illustrates the computation of the HPD overlap. We compute high-density regions for the predictive distribution (top) and the normalized one-spike likelihood (middle), then divide the volume of their intersection by the volume of the smaller region (bottom).

Predictive checks

The HPD overlap compares two distributions of the state: one based on the accumulated past information and the other based on the isolated information from the likelihood of the most recent spiking observation. Another approach is predictive checking, which compares the observation itself to a predictive distribution of that observation estimated from the model. This approach has been used before in other contexts ([11–13]) and has the advantage that it allows for the computation of p-values for individual observations, which are easy to interpret [14, 15].

Predictive checks for spiking data can be challenging to implement, since within sufficiently short intervals, a spike is typically less probable than no spike. One possible approach is to consider spike patterns over longer fixed intervals in which multiple spikes are expected. Here, we take a simpler approach of focusing only on spike times and measuring the likelihood that an observed spike comes from a particular neuron (for spike-sorted models) or has a particular set of waveform features (for clusterless models).

Using the event notation introduced above, the relevant observation distribution is the distribution of the spike's mark for an event drawn from the model-implied marked point process. Because the diagnostic is evaluated once for each spike event, its predictive reference distribution corresponds to selecting an event rather than selecting a time bin that contains one or more events. For a spike-sorted model, let $\Lambda(x_k) = \sum_{d=1}^{C} \lambda_d(x_k)$ denote the total expected population count in the bin. Under the one-step predictive distribution P_k, the expected number of events associated with a state region dx_k is proportional to $\Lambda(x_k)P_k(x_k)\,dx_k$. The latent state associated with a randomly selected predicted event therefore has the event-weighted distribution

$$P_k^{\text{event}}(x_k) = \frac{\Lambda(x_k)P_k(x_k)}{\int \Lambda(u)P_k(u)\,du}. \tag{3}$$

At states where $\Lambda(x_k) > 0$, the probability that the event came from neuron c is

$$\pi_c(x_k) = \frac{\lambda_c(x_k)}{\Lambda(x_k)}. \tag{4}$$

States where $\Lambda(x_k) = 0$ receive zero mass under P_k^{event}. Combining the event-weighted state distribution with the conditional cell-identity distribution gives the predictive probability of cell identity:

$$f_{\text{pred}}(c) = \int \pi_c(x_k)P_k^{\text{event}}(x_k)\,dx_k = \frac{\int \lambda_c(x_k)P_k(x_k)\,dx_k}{\sum_{d=1}^{C}\int \lambda_d(x_k)P_k(x_k)\,dx_k}. \tag{5}$$

For a clusterless model, let $\lambda(y, x_k)$ be the expected event count per bin and unit mark volume at waveform mark y, so that $\Lambda(x_k) = \int \lambda(v, x_k)\,dv$ is again the total expected event count in the bin. The corresponding predictive mark density is

$$f_{\text{pred}}(y) = \frac{\int \lambda(y, x_k)P_k(x_k)\,dx_k}{\int \Lambda(x_k)P_k(x_k)\,dx_k}. \tag{6}$$

Equation (5) is the discrete spike-sorted form of Eq. (6). Let $\widetilde{y}_{k,j}$ denote a replicated mark distributed according to f_{pred}. We compute the probability that the observed spike is at least as likely as this replicate:

$$p_{k,j} = \Pr_{\widetilde{y}_{k,j} \sim f_{\text{pred}}} \left(\log f_{\text{pred}}(y_{k,j}) \geq \log f_{\text{pred}}(\widetilde{y}_{k,j}) \right) . \tag{7}$$

For the spike-sorted models considered here, $\widetilde{y}_{k,j}$ takes one of a finite set of values—the identities of the recorded neurons. We compute Eq. (7) exactly as

$$p_{k,j} = \sum_{\substack{c \in \{1,\ldots,C\}: \\ f_{\text{pred}}(c) \leq f_{\text{pred}}(y_{k,j})}} f_{\text{pred}}(c), \tag{8}$$

which sums the predictive probabilities of all cell identities at least as unlikely as the observed identity. The same probability could instead be approximated by Monte Carlo for a continuous or otherwise intractable mark space: draw x_k from the event-weighted distribution P_k^{event}, draw a replicated mark conditional on that state and an event, and compute the fraction whose predictive density is less than or equal to that of the observed mark. Equivalently, one may simulate from the full state-dependent point process and retain the marks of generated events. Because $p_{k,j}$ ranks the observed spike against this predictive distribution of possible spikes, we refer to it as the *rank-based predictive p-value*.

Fig 2b illustrates the sampling interpretation of a predictive p-value. Values of the state are sampled from the event-weighted predictive distribution (top), each of which is used to sample a mark that could accompany the event (middle). The predictive density of the observed mark is then ranked against the densities of the replicated marks (bottom). In the schematic, the total expected event count is constant across state, so the event-weighted state distribution equals the one-step prediction distribution. Although this diagram shows the Monte Carlo construction, the spike-sorted results in this paper use the equivalent exact finite sum. High p-values suggest that the observed spike is consistent with the predictive distribution of spikes and a p-value near zero suggests that the observed spike is unexpected based on the predictive distribution computed from the past spiking and the model.

While the HPD overlap ratio explicitly assesses our definition of consistency in the state space, the predictive check probes a related notion in the observation space through the marginalization in Eq. (6). When the observation model is sufficiently informative about the state, we expect large p-values when the predictive distribution and likelihood match, but also when either distribution is narrower than the other as long as they point to overlapping regions of the state space. This correspondence is not guaranteed: marginalization can map distinct state distributions to similar mark distributions, so a high predictive p-value does not by itself establish that the state distributions overlap. In the common case that the predictive distribution of the state is narrow relative to the likelihood, the predictive distribution of observations will be concentrated on the spikes that occur at the confidently predicted location, and the observed spike will have appreciable likelihood at that location. The observed spike is among those that the model anticipates and should therefore lead to a high p-value. In the less common case that the likelihood is narrow relative to the predictive distribution of the state, the predictive distribution of the observations will also be broad and will assign appreciable probability to a wide range of spikes. The observed spike is likely to fall within this anticipated range, so the p-value should also be high. A low p-value tends to arise when the observed spike is specifically unlikely relative to what the model anticipates, which corresponds to the likelihood concentrating in a region the accumulated past deems improbable.

KL divergence

Finally, one of the most well-known measures of discrepancy between probability distributions is the Kullback–Leibler (KL) divergence [16], defined in this case as the expected excess logarithmic loss incurred when the normalized one-spike likelihood is used to approximate the predictive distribution, with the expectation taken with respect to the predictive distribution:

$$D_{\mathrm{KL}}(P_k \,\|\, Q_{k,j}) = \int P_k(x_k) \log\left(\frac{P_k(x_k)}{Q_{k,j}(x_k)}\right) dx_k, \tag{9}$$

where P_k and $Q_{k,j}$ are the distributions defined above. This definition matches the per-spike likelihood used for the HPD overlap and in the likelihood rows of Figs 3 and 4.

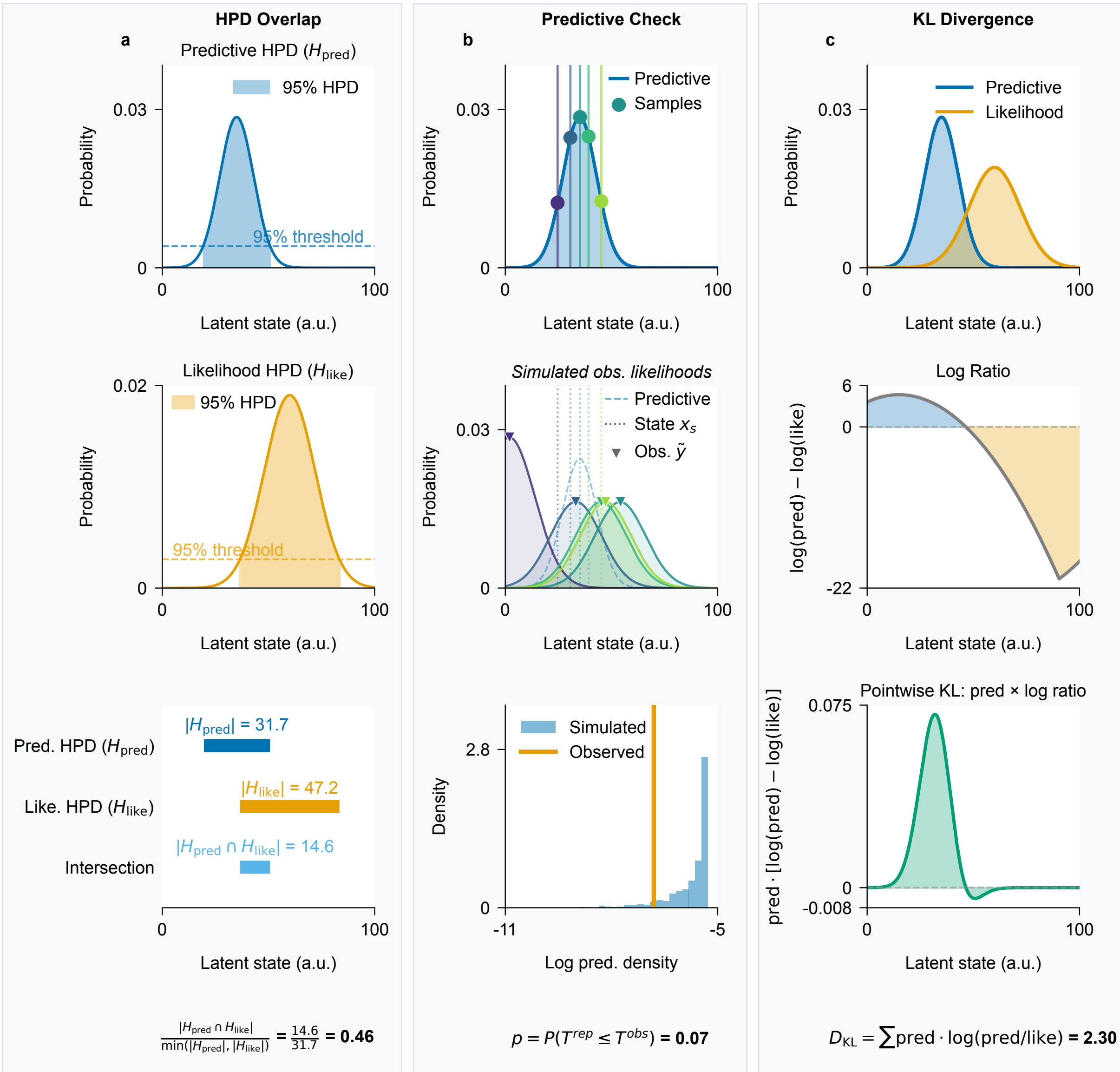

Fig 2. Computation of the three local goodness-of-fit metrics. **(a)** For HPD overlap, high-density regions are computed for the predictive distribution (top) and the normalized one-spike likelihood (middle). The volume of their intersection is then divided by the volume of the smaller region (bottom). **(b)** The conceptual sampling construction for the predictive check: state values are sampled from the event-weighted predictive distribution (top), and each sampled state is used to generate a mark that could accompany the event (middle). The total expected event count is constant across state in this schematic, so the event-weighted and ordinary one-step predictive distributions coincide. The metric is the probability that the log predictive density of the observed mark is greater than or equal to that of a replicated mark (bottom); for the finite spike-sorted observation spaces analyzed here, this probability is evaluated exactly rather than by sampling. **(c)** The KL divergence compares the predictive distribution (blue) and the one-spike likelihood (orange) (top) by computing the difference in their log density (middle) and computing its expectation with respect to the predictive distribution (bottom).

Fig 2c illustrates the computation of the KL divergence. The top panel shows the predictive distribution and normalized one-spike likelihood as functions of x_k. The middle panel shows their pointwise log-density ratio, $\log\big(P_k(x_k)/Q_{k,j}(x_k)\big)$. The bottom panel shows this ratio weighted by $P_k(x_k)$, which, when integrated, gives the value of the KL divergence.

The KL divergence is the most familiar way to compare two distributions, and we include it as a natural point of reference. It is an asymmetric measure, meaning that $D_{\mathrm{KL}}(P \parallel Q) \neq D_{\mathrm{KL}}(Q \parallel P)$. We could also compute the KL divergence from the likelihood to the predictive distribution, $D_{\mathrm{KL}}(Q \parallel P)$, or the average of these two divergences, $\frac{1}{2}(D_{\mathrm{KL}}(P \parallel Q) + D_{\mathrm{KL}}(Q \parallel P))$, but both alternatives lead to a common issue, namely that the KL divergence can become arbitrarily large when its first argument places appreciable probability where its second argument assigns very little. Thus, $D_{\mathrm{KL}}(Q \parallel P)$ can become large when an uninformative spike produces a broad likelihood. Choosing $D_{\mathrm{KL}}(P \parallel Q)$ avoids that particular behavior but raises a complementary one: the divergence will grow large whenever the predictive distribution is broad relative to the likelihood, which can occur, for example, whenever there are sparsely firing, informative spikes. We will see in the simulation and data analysis below that this flags many spikes, even when their information is consistent with the accumulated information from past spiking.

Simulation study

Validating local goodness-of-fit metrics on real neural data is fundamentally limited: when the truth is unobserved, a metric that flags a moment of model failure cannot be distinguished from one that flags a moment of unusual neural activity. We therefore first evaluate the three diagnostics—HPD overlap, the rank-based predictive p-value, and KL divergence—in a controlled benchmark in which each perturbation is known by construction. The benchmark simulates a hippocampal place-cell population on a one-dimensional track and runs a Bayesian state-space decoder over the full session. Three windows deliberately perturb either the generative process or the decoder's assumptions; two additional controls test unusual but internally consistent neural activity. Together, these conditions distinguish model misfit that leads to decoding errors from differences in distribution shape that do not imply conflicting state information. This relates to our original goal of identifying local periods where the information from the current spiking is inconsistent with the aggregated information from our model based on past spiking.

Generative model

The simulation applies a Bayesian filter to a 32-second session containing three misspecification windows, three clean-recovery windows, and two internally consistent controls.

Track and observation model. The state space is a one-dimensional linear track of length $L = 100$ arbitrary units (a.u.) discretized at unit spacing, $x \in \{0, 1, \ldots, 100\}$. The simulated population consists of eleven place cells with Gaussian place fields centered at $\mu_c \in \{0, 10, \ldots, 100\}$ and shared standard deviation $\sigma_{\mathrm{pf}} = 10$ a.u. The expected spike count of cell c per 1 ms time step at position x is

$$\lambda_c(x) = \alpha\, \phi\Big(x \mid \mu_c, \sigma_{\mathrm{pf}}^2\Big)\,, \tag{10}$$

where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density and $\alpha = 5$ is a rate-scale parameter. The peak expected count is $\alpha/(\sigma_{\mathrm{pf}}\sqrt{2\pi}) \approx 0.20$ spikes per step, corresponding to a peak rate of approximately 200 Hz. Spike counts are drawn independently across cells at each step, $y_{c,t} \sim \mathrm{Poisson}(\lambda_c(x_t))$. Except during the replay control described below, x_t is both the animal's physical position and the position represented by the simulated ensemble.

Five additional cells form a sparse population of narrow fields clustered near $x = 30$ a.u. (centers spread over 28.5–31.5 a.u., each with standard deviation 2 a.u.). Each fires as an independent Poisson process. Its per-cell peak rate is 0.01 Hz during the baseline and recovery periods and 1 Hz during the final sparse-population control; with five cells the aggregate rate stays low (a few spikes per second), so the spikes remain temporally isolated. The decoder uses these same epoch-specific rates, so recruiting these cells does not itself create an observation-model mismatch.

Position dynamics. During baseline windows, the animal's position evolves as a Gaussian random walk with step standard deviation $\sigma_{\text{pred}} = 0.5$ a.u. The trajectory is reflected at 0 and 100 a.u. throughout the session, starts at $x_0 = 0$, and is continuous across phases: the final position of each phase becomes the initial position of the next.

Phase schedule. The 32-second simulation ($T = 32{,}000$ steps) is partitioned into eight contiguous phases: an opening baseline, three misfit windows, three clean-recovery interludes, and a final sparse-population control. Each misfit window perturbs the observation model $p(y_k \mid x_k)$ or the state-transition model $p(x_k \mid x_{k-1})$ by altering either the generative process or the decoder's assumptions:

Remap: During steps 6,000–10,000, the decoder-side observation model is perturbed. The decoder evaluates per-spike likelihoods against a globally remapped set of place-fields: the eleven place-field centers are scrambled by a fixed random permutation (a derangement in which every field is displaced by at least three positions), modeling global remapping to statistically independent locations. Both the decoder's posterior update and its per-spike diagnostics use this remapped likelihood—the diagnostics are evaluated against the model the decoder actually uses, not the true generative fields—so the misfit surfaces from the scramble's spatial incoherence rather than from any oracle knowledge of the correct fields. Spike generation is unchanged.

History-dependent firing: During steps 14,000–18,000, the generative-side observation model is perturbed. Spiking is generated with a 1 ms hard refractory period followed by a 2–10 ms burst window with a threefold rate increase, matching the rough phenomenology of CA1 pyramidal-cell refractory and burst windows; the decoder still assumes independent Poisson firing. The per-spike spatial likelihood is unchanged, so the misfit lives in the spike-train temporal joint distribution.

Drift: During steps 22,000–26,000, the generative-side state-transition model is perturbed. The trajectory is generated with persistent velocity, $v_t = 0.88\, v_{t-1} + \eta_t$ and $x_t = x_{t-1} + v_t$ with

$\eta_t \sim \mathcal{N}(0, \sigma^2_{\text{pred}})$; the decoder continues to assume a memoryless random walk with step standard deviation σ_{pred}.

Sparse-population control. During steps 30,000–32,000, the animal is immobile at $x = 30$ a.u. and the ordinary place-cell ensemble becomes quiet, as can occur during immobility. The clustered sparse-population cells increase to their 1 Hz peak rate and fire intermittently. Unlike the misfit windows, this is an internally consistent control: the decoder knows both the quiet ensemble rate and the active sparse-population rates, and decodes with its usual random-walk motion prior. Because that prior expects the animal to move, in the intervals between isolated spikes the predictive distribution gradually spreads; each sparse-population spike then supplies a narrow likelihood centered inside that prediction. The two distributions therefore remain consistent by the HPD-overlap and predictive-check criteria, while their large difference in concentration can elevate KL divergence.

The three intermediate clean-recovery windows (steps 10,000–14,000; 18,000–22,000; 26,000–30,000) restore both the generative process and the decoder to the baseline configuration, providing out-of-sample well-specified comparison periods. They are summarized in Fig. 3b but are not used to set the diagnostic thresholds, which are derived from the opening baseline. During the last second of the final recovery, the animal moves gradually to the fixed location so the sparse-population control begins without an abrupt position jump. With this schedule fixed, each diagnostic can be evaluated against the known model faults and the two internally consistent controls.

Replay control. Embedded in the second clean-recovery window (steps 19,000–21,000) is a replay event. For this control, we distinguish the animal's physical position, z_t, from the internally represented position, x_t, that drives the simulated spiking. The animal is immobile, so z_t remains fixed, while x_t follows a coherent trajectory that sweeps out and back across the track. The decoder's observation and motion models are consistent with the replayed spikes, so it faithfully tracks x_t—its decoded position departs markedly from the animal's fixed physical position z_t while the predictive distribution and each spike's likelihood remain in agreement throughout the sweep. Because the diagnostics assess the internal consistency of the decoded state rather than its correspondence to physical position, this large decoded-versus-physical divergence is correctly left unflagged: it reflects a change in what the ensemble represents, not a failure of the model. The replay steps are excluded from the pooled well-specified comparison column and are scored as their own column in Fig 3b.

Simulation results

Fig 3a shows a realization of the simulation, the decoding results, and the three local goodness-of-fit metrics. The top panel shows the physical position of the simulated rat in magenta and the predictive decoding distribution in black. The regions with white backgrounds are the clean-recovery periods; colored backgrounds indicate the three model-misfit periods and two controls. The second panel shows the mean normalized per-spike likelihood as a function of position in each spike-containing time bin; normalized likelihoods are averaged when multiple spikes share a bin. The third panel shows the spiking from each of the sixteen simulated neurons. Since these are sorted by their place field centers, the pattern of spiking aligns with the likelihood as a function of position. Note that the remapped period is the one in which the predictive distribution and the spike likelihoods most clearly disagree about the animal's location. During the replay control, by contrast, the predictive distribution departs strongly from the physical position (the decoder tracks the replayed trajectory) yet continues to agree with the spike likelihoods, so the divergence is benign. The sparse-population control is visibly different: after the ordinary ensemble falls silent, isolated spikes appear only in the clustered sparse-population cells.

The lower three panels of Fig 3a show, from top to bottom, the values of the HPD overlap, the rank-based predictive p-value (as $-\log(p)$) and KL divergence for each spike. In each panel a horizontal line marks the threshold used to flag poor fit. Because low HPD overlap indicates disagreement, a spike is flagged when its value is at or below the 1st percentile of values from the opening baseline (steps 0–6,000) pooled across 100 independent simulation realizations. For the rank-based predictive p-value, a spike is flagged when $p \leq 0.05$ (equivalently, $-\log(p) \gtrsim 3$), a fixed cutoff rather than a baseline-derived threshold. Because high KL divergence indicates disagreement,

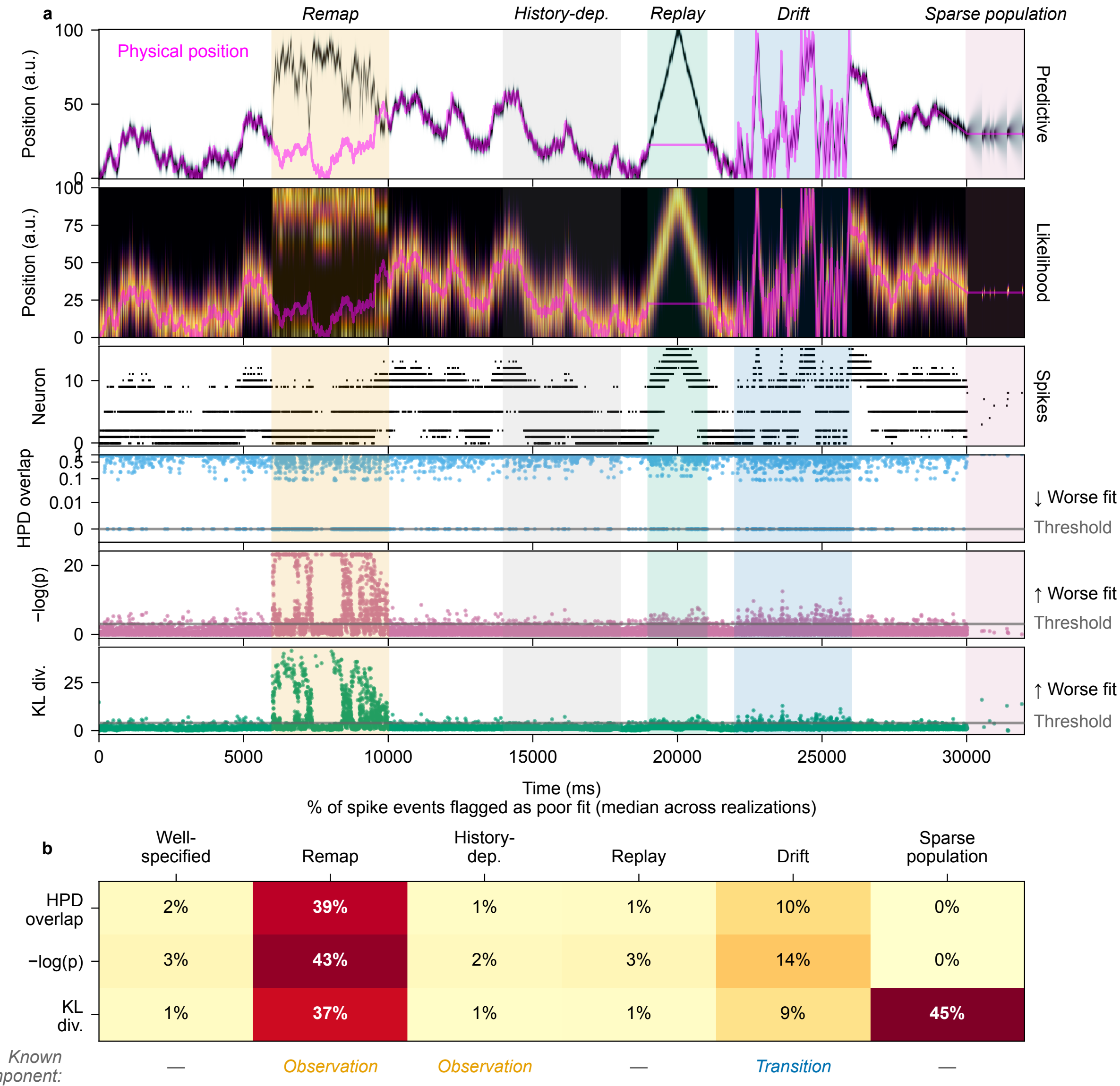

Fig 3. Per-spike goodness-of-fit diagnostics for the simulation study. **(a)** The top row shows the predictive distribution over time and position, with the animal's physical position in magenta and the three misfit periods and two controls shaded. The second row shows the mean normalized per-spike likelihood in each spike-containing bin (averaged when spikes share a bin), with physical position again in magenta. The third row shows the spike raster with neurons sorted by place-field peak position. The bottom rows show HPD overlap, the rank-based predictive p-value, and KL divergence for each spike event, with thresholds for poor fit; HPD overlap uses a symmetric-log y-axis to resolve values near zero. **(b)** The percentage of spike events flagged in each phase, summarized by the median across 100 independent realizations. The well-specified clean-recovery windows are pooled into one column, and each control has its own column. Shared HPD and KL thresholds are derived from opening-baseline values (steps 0–6,000) pooled across all realizations; the predictive p-value uses the fixed cutoff of 0.05. A row beneath the heatmap identifies the perturbed model component—the observation model (*Observation*) or state-transition model (*Transition*)—and uses a dash for well-specified and control phases.

a spike is flagged when its value is at or above the 99th percentile of values from the same pooled opening-baseline distribution. Pooling across realizations stabilizes the baseline-derived HPD and KL thresholds, which would otherwise be noisy when estimated from a single run.

We see that all three metrics agree that there is a lack of consistency during the remapping period, where the likelihood from the remapped cells is in a different region of the environment than the model estimate based on the incorrect place field estimates. During the period when the history dependence of spiking is misspecified, the estimated decoding distribution becomes narrower but remains consistent with the spike likelihoods, and none of the three metrics flag this period as leading to incorrect estimates. The replay control is likewise unflagged, though for a different reason: although the decoded position sweeps far from the animal's physical location, the predictive distribution tracks the replayed trajectory and remains consistent with each spike's likelihood, so all three metrics stay at their baseline levels. During the period when the decoder misses a drift term, the actual movement trajectory has notably increased velocity. The decoder only partially keeps up, so its predictive estimates lag behind the physical location; this lag biases the predictive spiking distribution toward neurons consistent with the lagged position. As a result, the diagnostics flag a modest percentage of spikes during the drift period—most clearly the rank-based predictive p-value—though far less than during the remapping period. During the sparse-population epoch, the ordinary ensemble is quiet and the prediction broadens naturally between intermittent spikes from the clustered cells. Each such spike concentrates inside the range of positions already supported by the prediction. HPD overlap and the rank-based predictive p-value therefore classify the two distributions as consistent, whereas KL divergence responds to their large difference in concentration. This familiar sparse-firing regime illustrates why distributional difference need not imply conflicting state information.

Fig 3b shows the percentage of spikes flagged by each metric in each period, summarized by the median across 100 independent realizations of the simulation. A spike is flagged whenever its diagnostic crosses the corresponding threshold in the direction of worse fit: at or below the 1st-percentile baseline value for HPD overlap, at or below the fixed 0.05 cutoff for the rank-based predictive p-value, or at or above the 99th-percentile baseline value for KL divergence. The median percentage flagged during the remapping period is substantial for all three metrics (37–43%), demonstrating that all three methods are able to identify inconsistent spiking information leading to incorrect state estimates; the percentage nonetheless varies from realization to realization because it depends on how much of the scrambled map the simulated trajectory visits. All three metrics flag only 1–2% of spikes during the history-dependence period. The replay control is flagged at a low rate—a median of 1–3% across the three metrics, comparable to the well-specified baseline—confirming that a large decoded-versus-physical-position divergence is not, by itself, registered as model misspecification. The drift period is flagged more often (9–14%), most notably by the rank-based predictive p-value, but still far less than the remapping period. In the sparse-population control, HPD overlap and the rank-based predictive p-value flag no spikes, whereas KL divergence flags a median of 45%. During the quiescent intervals between isolated sparse-population spikes the decoder's estimate receives little new information and its prediction broadens; each spike then arrives as a sharply localized, spatially informative observation whose location remains well inside that broad prediction. HPD overlap and the predictive check therefore register the spike and the prediction as consistent, while KL responds to their large difference in concentration. This familiar sparse-firing regime—the kind that arises during immobility—illustrates that a distributional difference between prediction and likelihood need not reflect conflicting information about the state.

Real hippocampal data analysis

Data collection and behavioral task

We applied the local goodness-of-fit metrics to units recorded from the rat hippocampus while the animal performed a spatial bandit task on a maze with three radially arranged Y-shaped foraging patches extending from a central point. A schematic of the maze is shown in the right inset in Fig 4. The details of this task and the data collection methods are described by Comrie et al. [17]. Briefly, the rat was trained to run between six reward ports located in three Y-shaped "patches", each with a different probability of reward. Reward probabilities changed periodically after a preset time or trial limit. Recordings were made with 22 tetrodes advanced to CA1; two additional tetrodes, one in each hemisphere, served as local references in the corpus callosum.

For the present analysis, we sorted the continuous 30 kHz neural signal into putative single units with MountainSort4 [18]. In the MountainSort4 pipeline, the signal was band-pass filtered during preprocessing and whitened; negative-going events were then detected on each tetrode at a threshold of 3 in whitened units, with a minimum inter-event interval of 10 samples (≈0.33 ms). Units were retained from automated curation only, without manual cluster curation. Because curation was automated, some units may contain multiunit activity or represent incompletely merged clusters; this does not affect our purpose here, which is to demonstrate the diagnostics rather than to characterize individual neurons.

For decoding, we linearized the animal's position by projecting it to the nearest point of a skeletonized track, discretized the resulting linear position into spatial bins of approximately 2 cm, and binned spikes at 2 ms resolution. We fit both decoders with `non_local_detector` version `0.6.10.dev214+g956fdccaf`. They used the same sorted-spikes kernel-density observation model, with positional bandwidth `position_std`= $\sqrt{12.5} \approx 3.54$ cm, and initialized position uniformly over the track-interior bins. The Continuous model assigned probability one to its sole dynamics mode and used a zero-mean Gaussian random-walk position transition with `movement_var`= $6.0\,\text{cm}^2$. The Continuous–Fragmented model initialized the Continuous and Fragmented modes with probabilities $(0.5, 0.5)$ and used the mode-transition matrix

$$\begin{pmatrix} 0.98 & 0.02 \\ 0.02 & 0.98 \end{pmatrix},$$

with rows and columns ordered as (Continuous, Fragmented). Its Continuous-to-Continuous position transition used the same random walk as the Continuous model; the other three mode transitions used a uniform distribution over track-interior position bins.

The Continuous–Fragmented decoder represented the joint state of discrete dynamics mode and position, with the same position-dependent observation model in both modes. For all three diagnostics, we marginalized its one-step predictive distribution over dynamics mode and compared the resulting distribution over position with one copy of the shared observation likelihood. The diagnostics for both decoders were therefore defined on the same position grid. Because the recording has no designated clean baseline period, we used fixed cutoffs—HPD overlap at or below 0.05 and a rank-based predictive p-value at or below 0.05—rather than the within-session baseline percentiles used in the simulation.

Data analysis results

Fig 4 compares decoding of these hippocampal spike data under two state-space decoding models. The first assumes that the latent state moves continuously through space, as expected if it represents only the animal's physical location. During immobility, however, hippocampal representations can jump to distant locations and reactivate in sequences resembling past movement trajectories [19, 20]. We therefore expect the Continuous model to exhibit local disagreement with the spiking data when the neural representation jumps. We compare this model with a Continuous–Fragmented model that permits discrete jumps in the hippocampal representation. We previously used a related model structure to capture replay and other nonlocal hippocampal representations [2].

Fig 4a shows a brief segment decoded with the Continuous model while the animal is immobile. The panels follow the organization of Fig 3a. During this candidate replay event, the decoded representation transitions rapidly among maze locations while the animal remains at rest, as indicated by the magenta position trace. The Continuous model cannot track these rapid transitions, and its predictive distribution lags behind the information provided by the likelihoods of the observed spikes. The HPD overlap and rank-based predictive p-value diagnostics flag many spikes, and several spikes also have elevated KL divergence; KL divergence values are not dichotomized in this analysis.

The contrast between the Continuous and Continuous–Fragmented models is shown in panels a and b of Fig 4. The likelihood of each spike is identical under the two models (second row), but the Continuous–Fragmented model decodes a rapidly moving state trajectory that the Continuous

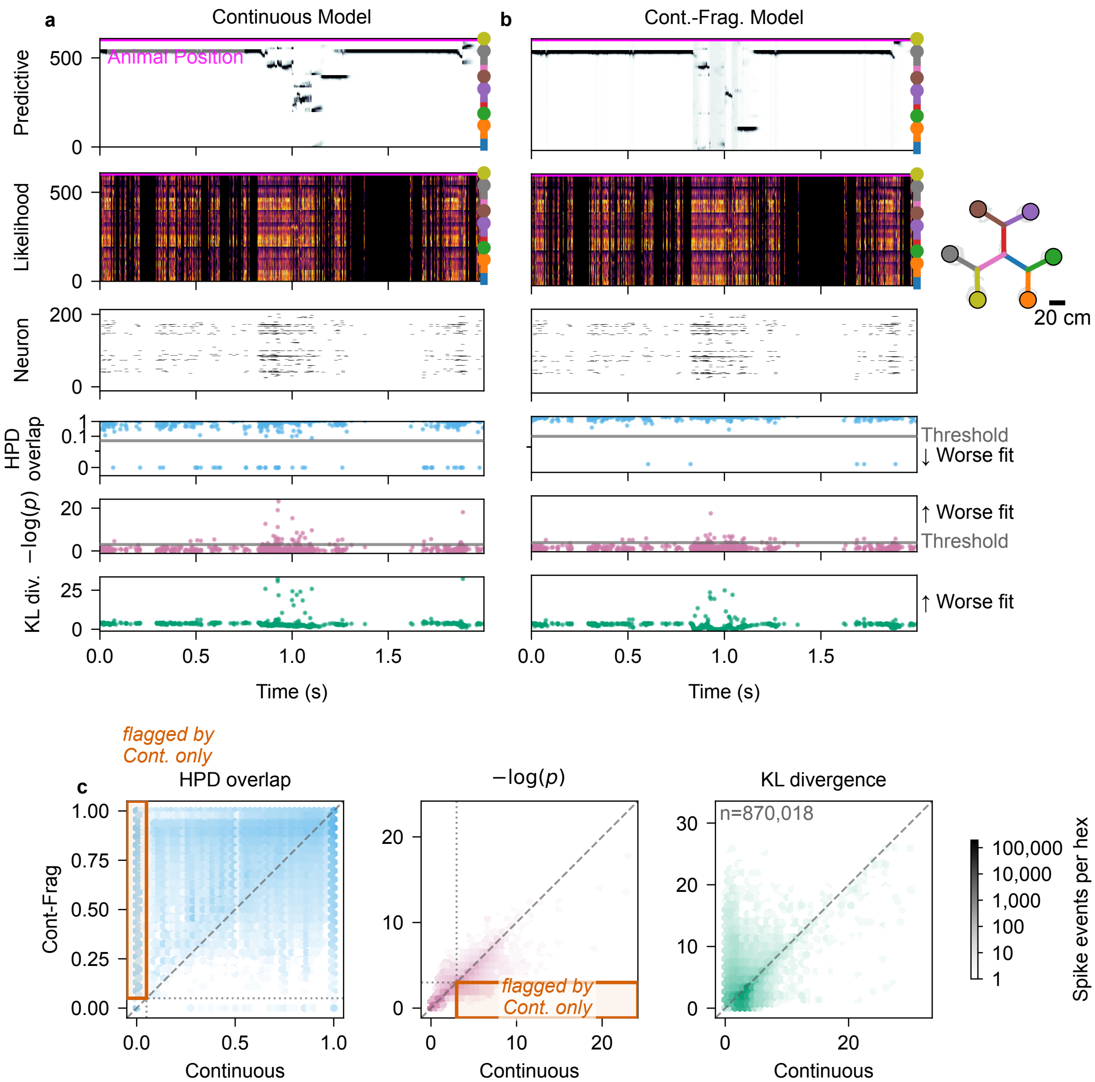

Fig 4. Per-spike goodness-of-fit diagnostics applied to a hippocampal recording containing 203 simultaneously recorded units during navigation on a multi-arm maze, using data from Comrie et al. [17]. Two state-space models—a Continuous model and a Continuous–Fragmented model—are evaluated on the same session. The unlettered track inset shows the two-dimensional maze layout, with arms drawn as colored segments, colored dots marking the six reward wells, the animal's trajectory overlaid in gray, and a 20 cm scale bar. **(a, b)** A detail window containing a candidate replay event, decoded with the Continuous and Continuous–Fragmented models, respectively. All rows share a common horizontal axis showing time in seconds relative to the start of the window. **Row 1, Predictive**: the one-step predictive distribution over linearized position, with the animal's physical position overlaid. The discrete dynamics mode of the Continuous–Fragmented model is marginalized so that both models are evaluated over the same position grid. **Row 2, Likelihood**: the mean normalized per-spike likelihood over position (as in Fig 3), identical under both decoders. **Row 3, Unit**: a spike raster with units sorted by place-field peak position. **Rows 4–6**: one marker per spike event for HPD overlap, $-\log(p)$, and KL divergence. Horizontal lines mark the flag cutoffs: HPD overlap and the predictive p-value are flagged at or below 0.05 (the latter shown as $-\log(0.05) \approx 3$); KL divergence has no fixed cutoff. The direction of poorer local fit is labeled at the right. **(c)** Pairwise comparisons of per-spike diagnostic values across the entire session, with the Continuous model on the x-axis and the Continuous–Fragmented model on the y-axis. The dashed line denotes identity. In the HPD overlap and $-\log(p)$ panels, dotted lines mark the corresponding flag thresholds, and the outlined box identifies spikes flagged by the Continuous model but not the Continuous–Fragmented model. Color intensity indicates the number of spike events on a shared logarithmic scale.

model cannot track. When the representation is assigned to the fragmented state, the predictive distribution becomes uniform or nearly uniform, allowing the likelihoods of subsequent spikes to produce a jump in the state estimate.

For HPD overlap (fourth row, blue), nearly all spikes flagged under the Continuous model are no longer flagged under the Continuous–Fragmented model. Some spikes also have lower $-\log(p)$ values under the Continuous–Fragmented model than under the Continuous model (fifth row, pink). Under the Continuous model, the predictive distribution can concentrate at the wrong location, making the observed spike very unlikely. During replay events, the Continuous–Fragmented model instead produces a broad predictive distribution. Although the observed spike can remain unlikely under this broad distribution, it is less unexpected than under the concentrated Continuous-model prediction. For KL divergence (bottom row, green), many spikes retain high values under the Continuous–Fragmented model. This pattern is consistent with the simulation results: predictive distributions that are broad relative to spike likelihoods produce larger KL values even when our operational criterion classifies those prediction–likelihood pairs as consistent.

Fig 4c shows hexagonally binned density plots comparing the per-spike values of each local goodness-of-fit metric under the Continuous and Continuous–Fragmented models; color indicates the number of spike events in each hexagon. The left panel shows HPD overlap in blue. Hexagons above the 45° line correspond to spikes with greater HPD overlap under the Continuous–Fragmented model. The dense vertical band at $x = 0$ and $y > 0$ (within the outlined box) represents spikes for which the Continuous model's predictive distribution has no overlap with the spike likelihood, whereas the Continuous–Fragmented model's predictive distribution has nonzero overlap. There is comparatively little density at $y = 0$ and $x > 0$, indicating that adding the fragmented state alleviates this disagreement for many spikes. Of the 18,337 spikes flagged under the Continuous model, 16,881 (92%) are no longer flagged under the Continuous–Fragmented model (*rescued*), whereas 173 spikes are flagged only under the Continuous–Fragmented model (*newly flagged*).

The center panel of Fig 4c shows the negative logarithm of the rank-based predictive p-value, $-\log(p)$, in pink. Hexagons below the 45° line correspond to spikes that are less unexpected under the Continuous–Fragmented model than under the Continuous model. The density band below this line extends beyond $x = 3$ (within the outlined box), corresponding to $p \leq 0.05$ because $-\log(0.05) \approx 3$. The corresponding region above the line, where the Continuous–Fragmented value exceeds 3, is substantially less dense. Thus, the rank-based predictive p-value identifies many spikes for which local disagreement is reduced under the Continuous–Fragmented model. Of the 33,954 spikes flagged under the Continuous model, 9,373 (27.6%) are rescued by the Continuous–Fragmented model, whereas 1,706 spikes are newly flagged.

The right panel of Fig 4c shows KL divergence values in green. Hexagons below the 45° line correspond to spikes with lower KL divergence under the Continuous–Fragmented model. A dense concentration of spike events lies below the line for Continuous-model values below 5. More strikingly, a region above the line contains Continuous-model values near 2 and Continuous–Fragmented values extending to approximately 20. As in the simulation, these values occur when the Continuous–Fragmented model produces a broad predictive distribution after estimating a fragmented state. KL divergence therefore aligns less closely with the nesting-tolerant consistency criterion used for the other metrics in this setting.

Discussion

State space modeling has become a powerful tool for describing the latent dynamics of neural systems. Validating these models is inherently difficult because latent states are unobservable and many models are purposefully simplified. To address this, we developed multiple local goodness-of-fit metrics that compare the instantaneous information received from each spike to the information carried forward from all preceding spiking activity by the state-space model. Rather than dismissing a model in its entirety, these metrics are able to assess the consistency between immediate observations and the information accumulated through the state-space structure, providing a more granular assessment of decoding reliability.

Two of the three metrics we examined, the HPD overlap ratio and predictive checks, met our criteria for assessing the consistency between the information in the predictive distribution and the likelihood. The HPD overlap offers a robust geometric measure of similarity, consistently maintaining lower error rates across various conditions. Predictive checks provide a direct statistical assessment through observation-level p-values, but are more computationally intensive. The KL divergence, in contrast, is computationally simple but is sensitive to distribution tails and lacks a clear upper bound, making it difficult to assess similarity. More broadly, KL divergence is only one member of the larger family of f-divergences, which also includes the total variation distance [21] and the Pearson χ^2 divergence [22]. Substituting another member of this family would emphasize different features of the prediction-likelihood mismatch, but because these measures all target distributional difference rather than the consistency we seek, KL's tendency to flag spikes that remain consistent with the model, as occurs whenever the predictive distribution is broad, would likely carry over to the family as a whole. The KL divergence (or any of these generalizations) may be useful in cases when the goal is to flag time points where the predictive distribution becomes uncertain relative to the information coming from individual spikes. When the goal is consistency in estimates of the state-process, we recommend the HPD overlap and predictive checks as primary diagnostic measures.

We illustrated the application of these metrics to simulated and real spike-sorted data from rat hippocampus. These same metrics naturally extend to other neural data types when incorporated into state space models. In particular, recent work has shown that clusterless neural spiking models can substantially reduce bias and variance in neural decoding and estimation problems when compared to spike-sorted models [23]. The metrics, as defined in the Methods section, naturally extend to clusterless models. In that case, a lack of fit for any particular spike suggests that a spike with that waveform was unexpected based on the accumulated information from the state space model up to that point in time.

These metrics make it possible to identify specific periods of time where a state-space model is more or less consistent with the spiking data. For example, we show that a model that includes a fragmented state that allows jumps in hippocampal representations provides a better fit to many spikes than a continuous-only model where these jumps are highly unlikely. More broadly, these approaches enable a feedback loop of model development and model assessment that could greatly improve the quality of our state-space models.

There are a number of ways we could extend these methods. For simplicity, we focused on comparing instantaneous information in the likelihood of each spike with aggregated information from the past contained in the one-step prediction density. A natural and powerful extension of these methods is obtained by replacing the one-step prediction density with any posterior density derived from the state-space framework. In particular, replacing the one-step prediction density with a smoothing density (e.g., from a fixed-interval smoother over the observation period) would allow us to leverage both past and future data to increase statistical power in identifying model misspecification.

This potential impact of these approaches does come with challenges, however. First, the computational burden of computing these metrics increases as the dimensionality of the state process increases. In all of our examples, we computed the KL divergence measure and the HPD overlap by partitioning the state space into a lattice and computing the likelihoods and prediction densities at each point in that lattice. As the dimension of the state increases, the size of that lattice will grow exponentially. We also evaluated the predictive checks exactly by summing over the finite set of recorded-neuron identities. For models with continuous or otherwise high-dimensional observation spaces, the predictive check could instead be approximated by sampling from the predictive state and observation distributions, avoiding enumeration of every observation outcome; whether this is more efficient will depend on the model and the required Monte Carlo precision. Sampling-based estimators could likewise improve the tractability of KL divergence and HPD overlap in moderate dimensions, although the number of samples required for accurate estimates will still grow with dimension. Another approach for dealing with this issue may be to compute these consistency metrics in lower-dimensional subspaces of the state space. Identifying subspaces that retain the statistical power of the original space is a topic for future exploration.

Another potential challenge relates to identifying time-scales for which to assess for lack of fit.

These methods allow us to focus on time-scales as short as individual spikes, and for our place cell decoding examples individual spikes are sufficiently informative that we are often able to detect local model misfit at this time-scale. For other neural systems, it may be the case that local model errors are more likely to be detected over longer time-scales. In this paper, we noted that these goodness-of-fit measures can be extended to longer time intervals, but determining what time-scales are most appropriate may require relying on prior knowledge about the system, or exploring the values of these metrics across multiple time-scales.

We believe that these local goodness-of-fit tools provide an important step toward making inferences from state-space neural models more reliable and interpretable. By pinpointing specific moments when the model assumptions break down, researchers can be more confident in their inferences. Ultimately, these measures empower us to refine our understanding of neural dynamics and deepen our understanding of how the brain encodes and processes information.

Data availability

The hippocampal recording analyzed here was originally reported by Comrie et al. [17] and is available on the Distributed Archives for Neurophysiology Data Integration (DANDI) Archive as dandiset 001942 (`https://dandiarchive.org/dandiset/001942`).

Code availability

All simulation, analysis, and figure-generation code is publicly available at `https://github.com/edeno/statespacecheck-paper`. [TODO: Add the archival DOI (e.g., Zenodo) for the tagged release used in this paper.]

Ethical statement

All animal procedures were approved by the Institutional Animal Care and Use Committee at the University of California, San Francisco (protocol AN174991).

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The authors declare that they have no competing interests.

Author contributions

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Software: SZ, ELD.

Formal analysis: SZ, ELD.

Investigation: SZ, UTE, ELD.

Data curation: AEC.

Writing – original draft: SZ, UTE, ELD.

Writing – review & editing: SZ, AEC, LMF, UTE, ELD.

Visualization: SZ, ELD.

Supervision: UTE.

Funding acquisition: UTE, LMF.

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